Smart Fan Speed Controller

Temperature + humidity
Fuzzy Logic and Genetic Programming PBL

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Problem

Manual speed adjustment depends on the user noticing changing conditions.

A temperature-only threshold ignores humidity and can switch abruptly.

Our model requests airflow from both inputs. It does not predict room cooling or moisture removal.

Objectives

Build a readable nine-rule Mamdani controller.

Visualize membership functions and explain a worked example.

Check coverage, bounds and invalid inputs across the declared domain.

Evolve expression trees to approximate simulated outputs.

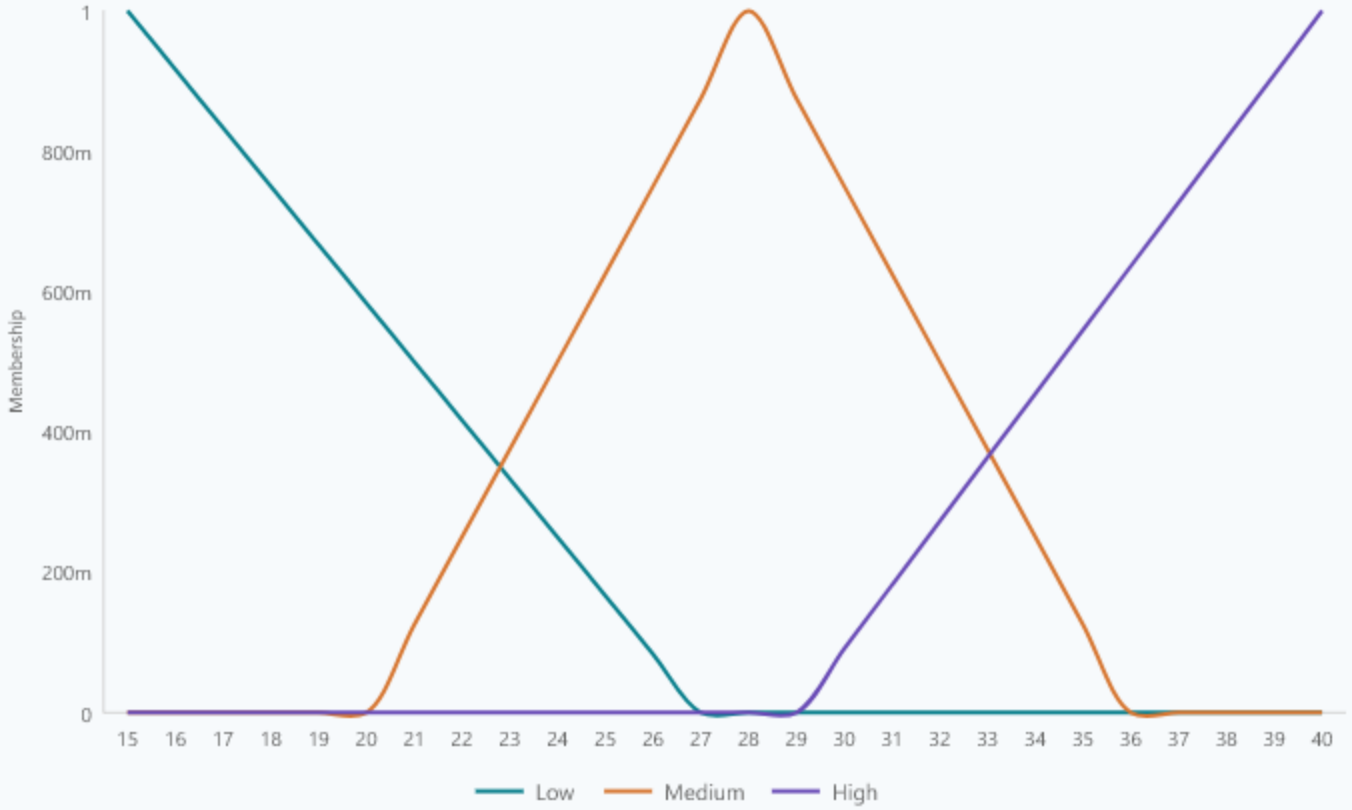
System Design

Variable	Domain	Meaning
Temperature	15–40 °C	Room condition input
Humidity	20–90% RH	Room condition input
Fan command	0–100%	Requested speed, not RPM

Validate → Fuzzify → Min rules → Max aggregation → Centroid

Software simulation. A future motor driver needs calibration.

Membership Functions



Temperature (°C)

Humidity: (20,20,55)
(35,60,85), (65,90,90)

Output: (0,0,50)
(25,50,75), (50,100,100)

Temperature sets: (15,15,27), (20,28,36), (29,40,40)

Nine Fuzzy Rules

Temperature / Humidity	Low	Medium	High
Low	Low	Low	Medium
Medium	Low	Medium	High
High	High	High	High

Example: IF temperature is Medium AND humidity is High, THEN fan command is High.

Implementation

AND: $\alpha_{ij} = \min(\mu\text{temperature}_i, \mu\text{humidity}_j)$

Aggregate: $A(z) = \max$ over rules of $\min(\alpha_{ij}, \mu\text{output}(z))$

Command: $\int z\, A(z)\, dz \,/\, \int A(z)\, dz$

NumPy integration with a 0.1-point output grid. Invalid inputs raise ValueError.

Worked Example: 30 °C, 80% RH

Input	Low	Medium	High
30 °C	0.000	0.750	0.091
80% RH	0.000	0.200	0.600

Medium output clips at 0.200. High output clips at 0.600.

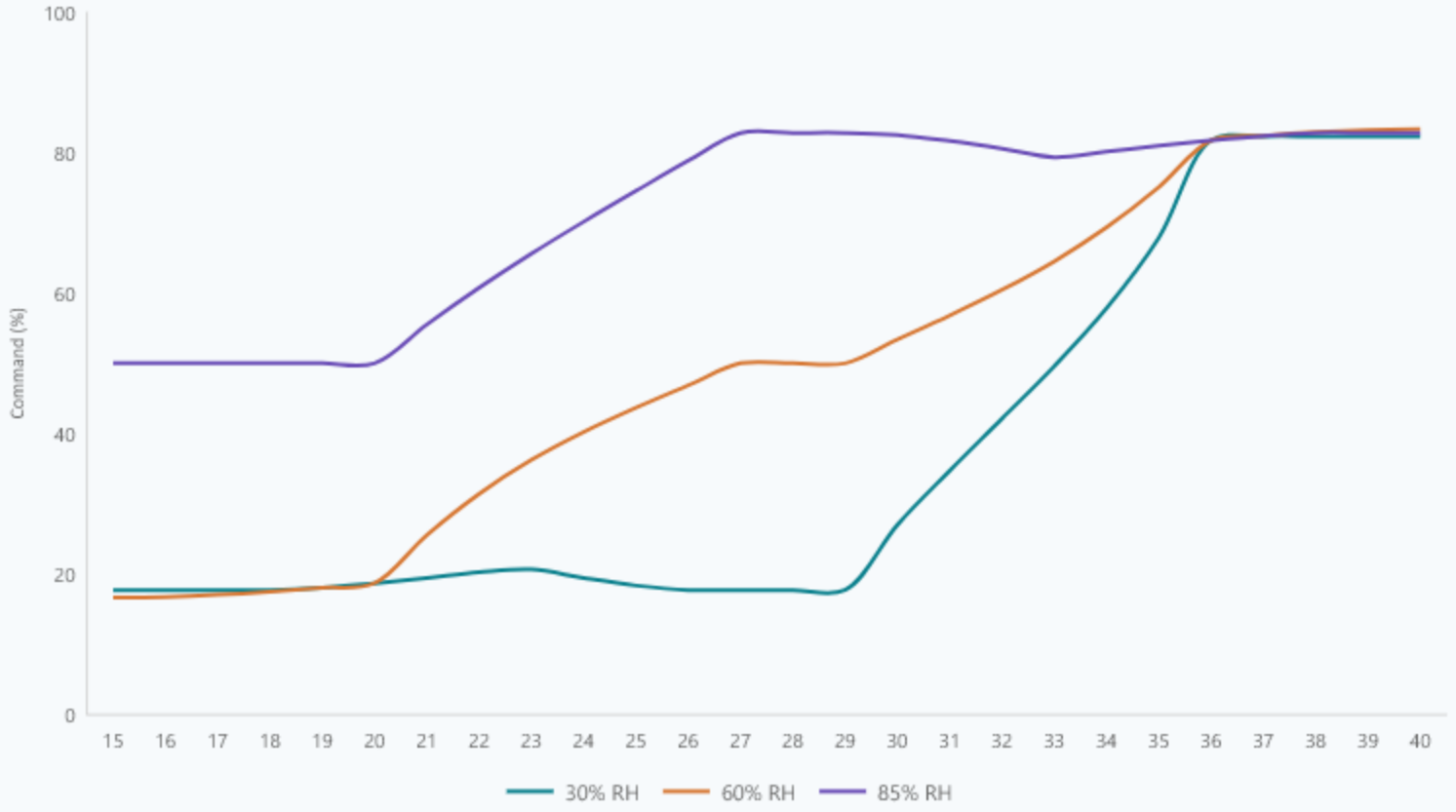
73.113% fan command

Simulation Results

Temperature (°C)	Humidity (%RH)	Command (%)
15	20	16.667
22	40	29.848
28	60	50
30	80	73.113
35	50	67.328
40	90	83.333

Low/high shoulder centroids limit this design to about 16.67–83.33%.

Analysis and Verification



Temperature (°C)

1,479 valid grid outputs

Local decreases:
1.358 points / 0.5 °C
2.387 points / 2.5% RH

No measured energy or comfort data

Genetic Programming

Expression trees with +, −, ×, min and max. Inputs normalize to 0–1.

100 trees, 40 generations, seed 71. Tournament selection and subtree variation.

85 training points + 36 held-out test points from an 11 × 11 grid.

Training RMSE: 9.215 points. Test RMSE: 9.431 points.

Approximation of synthetic fuzzy labels. The main controller remains the nine-rule model.

Conclusion and Next Steps

The controller covers the declared domain and explains its commands.

Local nonmonotonic behavior and GP approximation error remain visible.

Next: calibrate rules, enforce monotonicity and validate a physical fan.

Method references: official scikit-fuzzy control example and DEAP GP tutorial.
Full URLs and experiment details appear in the report and speaker notes.